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MILITARY SPECIFICATION

LATHES, SPEED, HORIZONTAL SPINDLE, SINGLE END, BENCH AND FLOOR MOUNTED

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bench and floor mounted, single end, horizontal spindle, speed lathes with equipment and accessories as described herein.

1.1 Classification. This specification covers speed lathes of the following types, classes and styles. The type, class, and style required for a specific procurement shall be as specified (see 6.2.1).

Type I - Hand Held Tool

Class 1 - Collet

Style A - Single spindle speed
Style B - Dual spindle speed
Style C - Infinitely variable spindle speed

Class 2 - Chuck

Style A - Single spindle speed
Style B - Dual spindle speed
Style C - Infinitely variable spindle speed

Type II - Machine Mounted Tool

Benefical comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis Tennessee 38114, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3419

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2. APPLICABLE DOCUMENTS

2.1 Military publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D C 20402.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA No. 79 - Electrical Standards for Metalworking Machine Tools

Z210.1 - Metric Practice (ASTM-E-380)

Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York 10018.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish a sample unit for first article inspection and approval (see 6.2.1 and 6.3).

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3.2 Design. The machine shall be new and one of the manufacturer's current models capable of performing secondary finishing operations such as polishing, filing, corner breaking, non-precision chamfering, and similiar operations. With Type I machines, the above operations shall be accomplished with hand-held tools. Type II machines shall be capable of being used to perform all of the above operations and, in addition, shall be capable of cut-off, turning, forming, and more precision chamfering operations by means of machine mounted tools. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U. S. Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the U. S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ANSI Z210.1.

3.2.1.1 Measuring and indicating device calibrations. When specified (see 6.2.1), measuring and indicating devices such as feed screw dials, scales, depth stops, carriage stops, dials indicators, pressure gauges, temperature indicators, and all other similar devices shall be graduated in the specified system (US or SI) of measurements. Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible without jeopardizing its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the materials originally selected for use on the machine. Reclaimed materials shall include iron, steel, copper, brass, aluminum, fiber products, plastics, and elastomers that have been collected from discarded solid, liquid, semi-solid, or gaseous waste such as garbage, refuse, or sludge.

3.2.3 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the

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guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.4 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.5 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective parts.

3.3.3 Fastening devices. All screws, pins, bolts and other fasteners shall be installed in a manner that prevents change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scales, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

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3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than 0.001 inch. Dial diameters shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a non-glare background.

3.3.8 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.3.9 Gears. All gears shall be machined from a material suitable for the intended purpose. All gears shall be heat treated by a process that will impart the hardness and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure or premature wear.

3.4 Components. The machine shall consist essentially of a mounting base, spindle, spindle drive, a work holding device, and their related controls. Any additional components necessary to the operation of the machine shall be furnished to the manufacturer's standard design and specifications.

3.4.1 Type I, Class 1, Style A.

3.4.1.1 Mounting base. The machine shall be designed for mounting onto a bench, pedestal, or steel cabinet. When specified (see 6.2.1), a pedestal or steel cabinet, as specified, shall be furnished. The pedestal or cabinet, when furnished, shall have provisions for secure attachment to the floor.

3.4.1.2 Spindle. The spindle shall be of the horizontal, single end type and shall be mounted on ball or roller bearings of a size and type suitable for operation at the specified speeds. The bearings shall be either permanently lubricated or shall have means for lubricating without the necessity of disassembling the spindle assembly. Bearings shall be sealed to prevent loss of lubricant or entrance of extraneous material. The spindle nose shall be designed to accommodate spring collets of the type and size specified in Table I, but shall also be capable of utilizing, as accessory equipment, a universal or independent jaw chuck of the size specified in Table I.

3.4.1.3 Spindle drive mechanism. The spindle shall be driven by a single speed motor providing the horsepower and spindle speed specified in Table II. The spindle may be either directly driven by the motor or through a belt drive mechanism.

3.4.1.4 Collet closer. The collet closer shall be operated by a hand lever which will permit rapid opening and closing of the collet. The collet closer shall be adjustable to accommodate variations in diameter of the workpieces.

3.4.1.5 Spindle brake. The machine shall have a mechanical or electrical brake capable of stopping the spindle, when operating at its rated speed, in not more than four seconds. Means of adjustment to compensate for brake lining wear shall be provided.

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3.4.1.6 Control. The control shall be a single lever, located convenient to the operator, and so arranged that it operates the collet closer, spindle brake, and motor switch.

3.4.2 Type I, Class I, Style B.

3.4.2.1 Mounting base. The machine shall be designed for mounting onto a bench, pedestal, or steel cabinet. When specified (see 6.2.1), a pedestal or steel cabinet, as specified, shall be furnished. The pedestal or cabinet, when furnished, shall have provisions for secure attachment to the floor.

3.4.2.2 Spindle. The spindle shall be of the horizontal, single end type and shall be mounted on ball or roller bearings of a size and type suitable for operation at the specified speeds. The bearings shall be either permanently lubricated or shall have means for lubricating without the necessity of disassembling the spindle assembly. Bearings shall be sealed to prevent loss of lubricant or entrance of extraneous material. The spindle nose shall be designed primarily to accommodate spring collets of the type and size specified in Table I, but shall also be capable of utilizing, as accessory equipment, a universal or independent jaw chuck of the size specified in Table I.

3.4.2.3 Spindle drive mechanism. The spindle shall be driven by a two speed motor providing the horsepower and spindle speeds selected from Table II and as specified (see 6.2.1). The spindle may be either directly driven by the motor or through a belt drive mechanism. When specified (see 6.2.1), the spindle shall have a spindle clutch which allows the spindle rotation to be stopped without stopping the motor. The spindle clutch shall smoothly engage, and shall be capable of transmitting full motor torque without slippage.

3.4.2.4 Collet closer. The collet closer shall permit rapid opening and closing of the collet and shall be adjustable to accommodate variations in diameter of the workpieces. The collet closer shall be actuated by one of the controls specified in 3.4.2.6.

3.4.2.5 Spindle brake. The machine shall have a mechanical or electrical brake capable of stopping the spindle, when operating at maximum spindle speed, in not more than four seconds. Means of adjustment to compensate for brake lining wear shall be provided.

3.4.2.6 Control. When the machine is equipped with a spindle clutch as described in 3.4.2.3, a single foot operated control shall operate the collet closer, spindle clutch and spindle brake. When a spindle clutch is not required, the collet closer shall be either hand operated or foot operated as specified (see 6.2.1). When specified (see 6.2.1), the collet closer, motor switch, and spindle brake shall all be operated by a single foot operated control.

3.4.3 Type I, Class I, Style C.

3.4.3.1 Mounting base. The machine shall be mounted on a steel cabinet type base with provisions for securely attaching the base to the floor.

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3.4.3.2 Spindle. The spindle shall be of the horizontal, single end type and shall be mounted on ball or roller bearings of a size and type suitable for operation at the specified speeds. The bearings shall be either permanently lubricated or shall have means for lubricating without the necessity of disassembling the spindle assembly. The bearings shall be sealed to prevent loss of lubricant or entrance of extraneous material. The spindle nose shall be designed primarily to accommodate spring collets of the type and size specified in Table I, but shall also be capable of utilizing, as accessory equipment, a universal or independent jaw chuck of the size specified in Table I.

3.4.3.3 Spindle drive mechanism. The spindle shall be driven by a single speed or two speed electric motor and an infinitely variable belt drive mechanism providing the spindle speed range and motor horsepower as selected from Table II and as specified (see 6.2.1). The machine shall have a spindle clutch which will allow the spindle rotation to be stopped without stopping the motor. The spindle clutch shall smoothly engage, and shall be capable of transmitting full motor torque without slippage.

3.4.3.4 Collet closer. The collet closer shall permit rapid opening or closing of the collet and shall be adjustable to accommodate variations in diameter of the workpieces. The collet closer shall be actuated by the control specified in 3.4.3.6.

3.4.3.5 Spindle brake. The machine shall have a mechanical or electrical brake capable of stopping the spindle, when operating at maximum spindle speed, in not more than four seconds. Means of adjustment to compensate for brake lining wear shall be provided.

3.4.3.6 Control. The control for the collet closer shall be a single hand-operated lever. The spindle clutch and brake shall be foot operated. A hand operated control for varying the spindle speed shall be provided.

3.4.4 Type I, Class 2, Style A.

3.4.4.1 Mounting base. The machine shall be designed for mounting onto a bench, pedestal, or steel cabinet. When specified (see 6.2.1), a pedestal or steel cabinet, as specified, shall be furnished. The pedestal or cabinet, when furnished, shall have provisions for secure attachment to the floor.

3.4.4.2 Spindle. The spindle shall be of the horizontal, single end type and shall be mounted on ball or roller bearings of a size and type suitable for operation at the specified speeds. The bearings shall be either permanently lubricated or shall have means for lubricating without the necessity of disassembling the spindle assembly. The bearings shall sealed to prevent loss of lubricant or entrance of extraneous materials. The spindle nose shall be designed to accommodate universal or independent jaw chucks of the size specified in Table I.

3.4.4.3 Spindle drive mechanism. The spindle shall be driven by a single speed motor providing the horsepower and spindle speed specified in Table II.

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The spindle may be either directly driven by the motor or through a belt drive mechanism.

3.4.4.4 Spindle brake. The machine shall have a mechanical or electrical brake capable of stopping the spindle, when operating at its rated speed, in not more than four seconds. Means of adjustment to compensate for brake lining wear shall be provided.

3.4.4.5 Control. The control shall be a single lever, located convenient to the operator, and so arranged that it operates the spindle brake and motor switch.

3.4.5 Type I, Class 2, Style B.

3.4.5.1 Mounting base. The machine shall be designed for mounting onto a bench, pedestal, or steel cabinet. When specified (see 6.2.1), a pedestal or steel cabinet, as specified, shall be furnished. The pedestal or cabinet, when furnished, shall have provisions for secure attachment to the floor.

3.4.5.2 Spindle. The spindle shall be of the horizontal, single end type and shall be mounted on ball or roller bearings of a size and type suitable for operation at the specified speeds. The bearings shall be either permanently lubricated or shall have means for lubricating without the necessity of disassembling the spindle assembly. The bearings shall be sealed to prevent loss of lubricant or entrance of extraneous materials. The spindle nose shall be designed to accommodate universal or independent jaw chucks of the size specified in Table I.

3.4.5.3 Spindle drive mechanism. The spindle shall be driven by a two speed motor providing the horsepower and spindle speed selected from Table II and as specified (see 6.2.1). The spindle may be either directly driven by the motor or through a belt drive mechanism. When specified (see 6.2.1), the spindle shall have a spindle clutch which allows the spindle rotation to be stopped without stopping the motor. The spindle clutch shall smoothly engage, and shall be capable of transmitting full motor torque without slippage.

3.4.5.4 Spindle brake. The machine shall have a mechanical or electrical brake capable of stopping the spindle, when operating at maximum spindle speed, in not more than four seconds. Means of adjustment to compensate for brake lining wear shall be provided.

3.4.5.5 Control. The machine shall have either a hand operated or a foot operated control as specified (see 6.2.1), for operating the spindle brake and motor switch. When the machine is equipped with a spindle clutch as described in 3.4.5.3, it shall have single foot control which operates the spindle brake and clutch.

3.4.6 Type I, Class 2, Style C.

3.4.6.1 Mounting base. The machine shall be mounted on a steel cabinet type base with provisions for securely attaching the base to the floor.

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3.4.6.2 Spindle. The spindle shall be of the horizontal, single end type and shall be mounted on ball or roller bearings of a size and type suitable for operation at the specified speeds. The bearings shall be either permanently lubricated or shall have means for lubricating without the necessity of disassembling the spindle assembly. The bearings shall be sealed to prevent loss of lubricant or entrance of extraneous materials. The spindle nose shall be designed to accommodate universal or independent jaw chucks of the size specified in Table I.

3.4.6.3 Spindle drive mechanism. The spindle shall be driven by a single speed or two speed electric motor and an infinitely variable belt drive mechanism providing the spindle speed range and motor horsepower as selected from Table II and as specified (see 6.2.1). The machine shall have a spindle clutch which will allow the spindle rotation to be stopped without stopping the motor. The spindle clutch shall smoothly engage and be capable of transmitting full motor torque without slippage.

3.4.6.4 Spindle brake. The machine shall have a mechanical or electrical brake capable of stopping the spindle, when operating at maximum spindle speed, in not more than four seconds. Means of adjustment to compensate for brake lining wear shall be provided.

3.4.6.5 Control. The machine shall have a foot operated control for operating the spindle clutch and brake and a hand operated control for varying the spindle speed.

3.4.7 Type II machines.

3.4.7.1 Mounting base. Unless otherwise specified, the mounting base shall be of the bench mounting type with provisions for secure attachment to a workbench. When specified (see 6.,2.1), the base shall be of the pedestal type with provisions for securely attaching to the floor.

3.4.7.2 Spindle. The spindle shall be of the horizontal, single end type and shall be mounted on ball or roller bearings of a size and type suitable for operation at the specified speeds. The bearings shall be permanently lubricated, or shall have means for lubrication without the necessity of disassembling the spindle assembly. The bearings shall be sealed to prevent loss of lubricant or entrance of extraneous materials. The spindle nose shall be designed to utilize both collets and chucks of the sizes specified in Table I.

3.4.7.3 Spindle drive. The spindle shall be driven by a single speed electric motor having the horsepower specified in Table II. The drive mechanism shall provide the number of spindle speed selections specified in Table II. Spindle speed selections shall be accomplished by shifting a belt on stepped pulleys.

3.4.7.4 Collet closer. The collet closer shall be a hand operated lever which will permit rapid opening and closing of the collet. The collet closer shall be adjustable to accommodate variations in diameter of work pieces.

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3.4.7.5 Bedway. The machine shall have a bedway for supporting a compound slide which shall accommodate a tool holder or grinder attachment. The bed shall be cast integrally with the spindle housing, and shall have a way that is machined, ground, or hand scraped in accordance with the manufacturer's standard commercial practice. The way may either be cast integrally with the bed and spindle housing assembly or may be a separate part which is securely attached to the bed. The length and width of the bedway shall conform to Table I.

3.4.7.6 Compound slide. The machine shall have a compound slide assembly consisting of a cross slide which rides on the bedway and a top slide which is mounted on the cross slide. The top slide and cross slide shall feed the tool in a direction square with the axis of the spindle. The top slide shall swivel on the cross slide through not less than a 360 degree arc and shall have graduations, in increments not larger than 1 degree, to indicate the angle to which the top slide is swiveled. Each fifth graduation shall be larger than the others and each tenth graduation shall be numbered. Means shall be provided for securely clamping the top slide to the cross slide at any swivel angle and for securely clamping the cross slide to the bed at any location on the bed. Both the cross slide and the top slide shall have manually operated feed screws that feed the tool the distances specified in Table I. The feed screws shall have dials conforming to 3.3.7. A tool post that will accept a 3/8 inch square high speed tool bit shall be furnished. The compound slide shall meet the performance rquirements 3.6.2.

3.4.7.7 Controls. The machine shall have a single, hand operated lever control for the motor switch and spindle brake. A separate hand control shall be provided to release the brake when the motor is stopped to allow the spindle to be manually rotated for inspection of the workpiece.

3.4.8 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to all requirements in this paragraph. The system shall conform to the requirements of ANSI/NFPA No. 79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit having a fused safety disconnect switch. The machine shall be initially wired for operation on the voltage specified (see 6.2.1). An identified terminal for grounding the machine when installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in Table 240-A of ANSI/NFPA No. 79 for the disconnect fuse rating.

3.4.8.1 Motors. Motors shall have ball or roller bearings. Continuous duty horsepower ratings shall conform to the reuirements of Table II. Unless otherwise specified (see 6.2.1), each motor shall meet the requirements of a drip-proof enclosure.

3.5 Sizes and capacities. The sizes and capacities of the machine shall conform to the requirements of Table I.

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Table I - Sizes and capacities.

	Type I						Type II
	Class 1			Class 2			
	Style A	Style B	Style C	Style A	Style B	Style C	
Number of spindle speed selections	1	2	Infinitely Variable	1	2	Infinitely Variable	3
Spindle speeds & motor horsepower	← See Table II →						
Spindle bore, inches	1/2	3/4	1	1/2	3/4	1	1-1/8
Collets:							
Type, series no.	3C	5C	5C	-	-	-	5C
Max. workpiece diam., inches	1/2	1-1/16	1-1/16	-	-	-	1-1/16
Chuck:							
Max workpiece diam., inches	*3	*6	*8	3	6	8	5
Type II machines:							
Swing over bed, inches	-	-	-	-	-	-	9
Bedway:							
Length (from spindle face to end of way), inches	-	-	-	-	-	-	10
Width, inches	-	-	-	-	-	-	2-1/2
Cross slide travel, inches	-	-	-	-	-	-	4
Top slide travel, inches	-	-	-	-	-	-	3

NOTE: Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure, or less, to the larger figure, or more.

* Denotes optional equipment furnished when specified (see 3.8 and 6.2.1).

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Table II - Spindle speeds and horsepower ratings of motors.

NOTE: The types, classes, and styles of speed lathes covered herein are available in the following spindle speeds and corresponding motor horsepowers. The spindle speed and corresponding motor horsepower required for a specific procurement shall be as specified (see 6.2.1).

Type I, Classes 1 and 2	Spindle Speed R. P. M	Corresponding Motor Horsepower Rating
Style A (single speed)	3450	1/4
Style B (dual speed)	Low High 570 1140 840 1725 1725 3450	* 1/4 or 1/2 * 1/3 or 1 * 1/2 or 1
Style C (infinitely variable speed)	100 to 600 100 to 1200 150 to 900 200 to 1200 200 to 2400 300 to 1800 400 to 2400 400 to 4800 600 to 3600 800 to 4800	1 or 2 1-1/2 or 3 2 or 3 3 or 5 1-1/2 or 3 2 or 3 3 or 5 1-1/2 or 3 2 or 3 3 or 5
Type II (3 speed belt drive)	Low 725 to 775 Med 1750 to 1850 High 2800 to 3200	1/2

* Horsepower ratings for dual speed machines are at high speed.

NOTE: Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure, or less, to the larger figure, or more.

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3.6 Accuracy and performance.

3.6.1 Types I and II. The spindle runout on all types, class, and style machines shall be not more than 0.0005 inch total indicator reading (TIR). There shall be no evidence of excessive vibration when operating the machine at the highest available spindle speed. Without chucks or collets installed, the brake shall stop spindle rotation, from the highest speed to a dead stop, in not more than four seconds.

3.6.2 Type II. Type II machines shall also meet the following additional performance requirements.

a. When facing a 5" diameter bar, as described in 4.5.3.1, the finished face shall be not more than 0.0005 inch concave or 0.0000 convex.

b. When turning a steep taper, as described in 4.5.3.2, the included angle of the finished taper shall be within 1/2 degree of the angle to which the compound slide is adjusted.

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

b. One set of collets to cover a range of sizes from 1/16 inch to maximum size shown in Table I in size increments of 1/32 inch (Type I, Class 1, Type II machines only).

c. One 3 jaw universal chuck, of the size specified in Table I (Type I, Class 2 and Type II machine only).

3.8 Optional equipment. The following optional equipment shall be furnished when specified (see 6.2.1).

a. Independent or universal jaw chuck, as specified, of the size specified in Table I.

b. Additional optional equipment as specified and fully described.

3.9 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.11 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the first article.

4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5.1 thru 4.5.3.2, where applicable, and the inspection in 4.6. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined to determine compliance with the reqirements in 3.2 through 3.5 and 3.7 through 3.11.

4.5 Tests.

4.5.1 Operational test (Type I and II). The machine shall be operated at no load for not less than 30 minutes during which time proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified. Spindle speeds and

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speed range, where applicable, shall be checked for conformance to Table II, and spindle braking time and vibration level shall be checked for conformance to the requirements of 3.6.1 during the operational test.

4.5.2 Spindle runout (Type I and II). The spindle runout shall be tested. The runout of the spindle nose shall conform to the rquirements of 3.6.1.

4.5.3 Type II machines. In addition to the above tests, Type II machines shall be subjected to the tests in 4.5.3.1 and 4.5.3.2.

4.5.3.1 Facing test. A 5 inch nominal diameter, SAE 2024 round aluminum bar of any suitable length shall be mounted in the chuck. With a high speed steel or carbide tool mounted in the tool holder, with the top slide swiveled at an angle, with the compound slide assembly clamped to the bedway, and with the spindle turning at a suitable speed, one or more trueing cuts shall be taken across the face of the workpiece using the cross feed screw. After the workpiece has been trued and rough faced, the top slide shall be fed in to produce a depth of cut not less than 0.002 inch deep. The finishing cut shall be taken using the cross feed screw. The flatness of the finished surface shall conform to 3.6.2.

4.5.3.2 Taper test. A 1 inch nominal diameter SAE 2024 round aluminum bar of any suitable length shall be mounted in the chuck or collet. The top slide shall be swiveled to the proper graduation for producing a 60 degree included angle. A 5 power magnifier shall be used to assure accurate alignment of the graduations. With the compound rest clamped to the bedways, the top slide shall be used to turn a 60 degree included angle point, extending from the center of the bar to the outer edge, on the free end of the workpiece. The included angle of the finished taper shall conform to the requirements of 3.6.2.

4.6 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing and marking shall conform to the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. Speed lathes covered by this specification are intended for secondary finishing operations such as filing, deburring, polishing, trimming and chamfering of metal parts.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not

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required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.

b. Type, class, and style required (see 1.2).

c. When a first article is required for inspection and approval (see 3.1, 4.2 and 6.3).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1.2).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.2).

g. Specify additional safety and health requirements (see 3.2.4)

h. Painting, if different (see 3.3.5).

i. When pedestal or steel cabinet base is required for Type I, Class 1, Style A, (see 3.4.1.1).

j. When pedestal or steel cabinet base is required for Type I, Class 1, Style B, (see 3.4.2.1).

k. Required spindle speeds and motor horsepower as selected from Table II (see 3.4.2.3).

l. Spindle clutch, if required (see 3.4.2.3).

m. Whether hand operated or foot operated collet closer is required (see 3.4.2.6).

n. Whether hand operated or foot operated control is required for spindle brake and motor switch (see 3.4.2.6).

o. Whether single foot operated control is required (see 3.4.2.6).

p. Required spindle speed range and motor horsepower as selected from Table II (see 3.4.3.3).

q. When pedestal or steel cabinet base is required for Type I, Class 2, Style A, (see 3.4.4.1).

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r. When pedestal or steel cabinet base is required for Type I, Class 2, Style B, (see 3.4.5.1).

s. Required spindle speeds and motor horsepower as selected from Table II (see 3.4.5.3).

t. Spindle clutch, if required (see 3.4.5.3).

u. Whether hand operated or foot operated control is required for spindle brake and motor switch (see 3.4.5.5).

v. Required spindle speed range and motor horsepower as selected from Table II (see 3.4.6.3).

w. Pedestal type base, if required (see 3.4.7.1).

x. Electrical equipment, if different (see 3.4.8).

y. Voltage requirements, if different and voltage of initial wiring (see 3.4.8).

z. Type of motor enclosure, if different (see 3.4.8.1).

aa. Standard equipment, if different (see 3.7).

bb. Optional equipment, if required, fully describe optional equipment required (see 3.8).

cc. Nameplate, if different (see 3.9).

dd. First article inspection, if different (see 4.2).

ee. Quality conformance inspection, if different (see 4.3).

ff. Preservation, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, wiring diagrams, and other instructions for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorporated into the contract.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The Contracting Officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article.

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6.4 Safety and health requirements. Paragraph 3.2.3 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the requiring agency is advised to analyze the existing hazards levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. If specific point-of-operation guarding is required, the requiring agency should specify the exact configuration of the guard required as in most cases the guard configuration is dependent on the size and configuration of the workpieces that will be machined. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(g).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force-99

Review Activities:

Army - AL
Navy - SH
DLA - GS

User Activities:
Navy - MC

Preparing Activity:
DLA - IP

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